

Name: _____

11.1 Circumference and arc length (exact and nearest tenth).

1. A circle has radius 13. Find its circumference.
2. Find the length of an 80° arc in a circle of radius 9.
3. Convert 300° to radians (exact).

11.2 Circle and sector areas.

4. A circle has radius 8. Find its area.
5. Find the area of a 135° sector in a circle of radius 8.

11.3 Polygon areas.

6. A kite has diagonals 22 and 9. Find its area.
7. Find the exact area of a regular hexagon with side 10.

11.4 Solids.

8. A dodecahedron has 12 faces and 20 vertices. How many edges?

11.5 Prisms and cylinders.

9. A rectangular prism is 11 by 6 by 4. Find its volume.
10. A cylinder has radius 7 and height 6. Find its volume.

- 11.** A prism's base is a triangle with base 10 and height 8; the prism is 9 tall. Find V .

11.6-11.7 Pyramids and cones.

- 12.** A square pyramid has base side 9 and height 14. Find its volume.
- 13.** A cone has radius 9 and height 8. Find its volume.
- 14.** A cone has radius 7 and height 24. Find its slant height and surface area.

11.8 Spheres.

- 15.** A sphere has radius 8. Find its surface area.
- 16.** Find the volume of the same sphere (nearest tenth).
- 17.** A hemisphere has radius 7. Find its volume (nearest tenth).

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- 18.** Find the area of a 150° sector in a circle of radius 12 (exact).
- 19.** A pyramid with a 4-by-15 base has volume 240. Find its height.
- 20.** A rocket model is a cylinder of radius 5 and height 6 topped by a cone of radius 5 and height 9. Find the total volume (exact and nearest tenth).